

PALACIOS, TX, USA

WMO: 722555

Lat: 28.725N

Lon: 96.254W

Elev: 4

StdP: 101.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12935

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	2.4	-6.2	2.2	6.2	-3.8	2.8	7.5	12.6	10.7	11.7	10.5	5.2	0	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	34.0	26.0	33.2	26.1	32.6	26.0	27.7	31.5	27.3	31.2	26.9	30.8	5.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.1	29.5	26.2	21.7	29.4	25.9	21.3	29.2	87.5	31.8	85.9	31.1	84.4	30.8	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.4	8.6	-2.6	36.5	2.0	1.8	-4.0	37.7	-5.2	38.8	-6.3	39.8	-7.8	41.1
			-3.8	28.2	1.9	0.6	-5.1	28.7	-6.2	29.1	-7.3	29.4	-8.7	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	12.9	14.9	18.0	21.5	25.4	28.3	29.2	29.3	27.0	22.9	18.0	13.9
	DBStd	6.92	5.01	5.02	4.34	3.33	2.48	1.64	1.29	1.55	2.46	3.90	4.78	5.09
	HDD10.0	71	28	16	4	0	0	0	0	0	0	0	4	19
	HDD18.3	593	177	114	61	12	1	0	0	0	0	10	64	154
	CDD10.0	4379	119	153	250	345	476	548	595	598	511	400	243	141
	CDD18.3	1861	9	18	49	107	219	298	337	339	261	152	53	18
	CDH23.3	18997	13	27	120	529	1922	3516	4246	4239	2738	1325	277	46
CDH26.7	7407	0	1	5	53	462	1431	1932	2018	1112	364	26	2	
Wind	WSAvg	4.7	4.8	5.0	5.2	5.5	5.4	4.8	4.4	4.3	3.8	4.3	4.4	4.5
Precipitation	PrecAvg	1192	85	62	65	73	120	126	97	95	177	106	83	76
	PrecMax	1684	261	175	235	235	336	395	311	234	603	338	234	208
	PrecMin	704	8	5	4	3	0	0	3	2	14	14	2	17
	PrecStd	285	56	44	57	59	95	103	82	69	148	78	63	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	25.7	27.3	29.8	31.7	34.3	34.4	35.4	35.0	32.4	28.7	26.3
		MCWB	19.3	20.5	20.8	20.5	24.9	25.5	25.6	25.9	25.4	24.8	23.8	21.9
	2%	DB	23.1	24.0	25.8	27.9	30.6	32.9	33.3	34.0	33.3	30.9	27.4	24.6
		MCWB	19.6	20.3	21.0	22.2	24.7	25.8	26.3	26.3	25.6	24.7	22.9	21.1
	5%	DB	22.0	22.9	24.7	27.1	29.9	32.2	32.7	33.3	32.4	29.9	26.2	23.1
		MCWB	19.1	19.7	20.8	22.4	24.5	25.9	26.3	26.3	25.4	24.2	21.9	20.4
	10%	DB	20.6	21.7	23.6	26.1	29.1	31.5	32.3	32.7	31.5	28.8	25.0	21.9
		MCWB	18.3	19.3	20.4	22.0	24.3	25.8	26.3	26.3	25.2	23.7	21.4	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	22.7	23.0	24.9	26.9	28.2	28.0	28.0	27.9	27.2	25.4	23.1
		MCDB	23.3	24.2	24.8	26.9	29.4	30.9	32.0	32.7	31.5	29.7	27.2	24.7
	2%	WB	20.8	21.7	22.5	24.2	26.2	27.4	27.4	27.6	27.2	26.4	24.2	22.2
		MCDB	22.1	23.0	24.2	26.3	28.7	30.7	31.7	32.1	30.6	28.8	26.0	23.7
	5%	WB	20.1	20.9	21.9	23.7	25.7	26.9	27.0	27.2	26.8	25.7	23.3	21.3
		MCDB	21.4	22.1	23.5	25.7	28.4	30.3	31.2	31.6	30.3	28.3	25.2	22.6
	10%	WB	19.0	20.1	21.3	23.1	25.2	26.6	26.8	26.8	26.3	25.0	22.4	20.4
		MCDB	20.1	21.2	23.0	25.1	28.0	30.0	30.8	31.1	29.9	27.7	24.3	21.6
Mean Daily Temperature Range	5% DB	MDBR	9.8	9.1	8.7	8.2	6.9	6.5	6.2	7.1	8.6	9.8	10.0	9.9
		MCDBR	9.5	8.5	7.7	7.9	7.0	7.0	6.8	7.7	9.0	9.1	9.2	9.7
		MCWBR	6.9	5.6	4.5	4.3	3.1	2.7	2.5	2.7	3.5	4.3	5.5	6.9
	5% WB	MCDBR	8.3	7.2	6.5	6.1	5.6	5.8	6.1	6.8	7.2	7.3	7.8	8.7
		MCWBR	6.9	5.6	4.4	3.8	2.9	2.7	2.5	2.7	3.3	4.3	5.8	7.0
Clear-Sky Solar Irradiance	taub	0.351	0.364	0.382	0.441	0.491	0.460	0.485	0.464	0.451	0.391	0.381	0.349	
	taud	2.497	2.469	2.409	2.244	2.136	2.288	2.232	2.293	2.296	2.457	2.428	2.517	
	Ebn at Noon	888	905	907	859	814	834	812	828	829	865	845	873	
	Edh at Noon	92	102	115	140	157	134	141	132	128	103	98	87	
All-Sky Solar Radiation	RadAvg	2.93	3.39	4.56	5.43	6.14	6.63	6.48	6.17	5.17	4.50	3.38	2.65	
	RadStd	0.31	0.55	0.41	0.47	0.40	0.41	0.58	0.36	0.47	0.47	0.37	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+108
(47)	Regional (34 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	+100

Nomenclature: See separate page